Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2024

Course : B.Tech | *Time : 03 Hours*
Code : 101617 | *Subject : Concrete Materials* | *Full Marks : 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 ***Choose the correct answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Who is the father of concrete technology?
(i) Royston Saint John
(ii) William Aspdin
(iii) Joseph Aspdin
(iv) Terzagi

(b) Which of the following are the ingredients of concrete?
(i) Binding material
(ii) Fine aggregate
(iii) Admixtures
(iv) All of the above

(c) High strength concrete is defined purely on the basis of _____
(i) Tensile strength
(ii) Good Aggregrates
(iii) Poor Aggregrates
(iv) Compressive strength

(d) What are Plasticizers?
(i) Which reduces water for workability
(ii) Which decreases workability at the same water content
(iii) Which oxidizes water for workability
(iv) Which adds water for workability

(e) When vibrators are used for compaction, the consistency of concrete depends upon the _____
(i) Type of mix
(ii) Placing conditions
(iii) Efficiency of vibrator
(iv) All of the above

(f) After the curing of 28 days, the concrete gains strength upto _____
(i) 40%
(ii) 80%
(iii) 60%
(iv) 100%

(g) A lower water cement ratio leads to _____
(i) Higher strength and durability
(ii) Lower strength but high durability
(iii) Higher strength but low durability
(iv) Lower strength and durability

(h) What is the compaction factor for low degree of workability?
(i) 0.78
(ii) 0.85
(iii) 0.92
(iv) 0.95

(i) Which cement is used in sewage and water treatment plants?
(i) Rapid Hardening Cement
(ii) Quick Setting Cement
(iii) Low Heat Cement
(iv) Sulphate Resisting Cement

(j) The presence of moisture in the sand _____ the volume of the sand.
(i) Increases
(ii) Shrink
(iii) Decreases
(iv) Contract

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Q.2 (a) What is light weight concrete? State the advantages of light weight concrete [7]
(b) Explain fibre types used in fibre reinforced concrete [7]

Q.3 (a) Discuss about the application of Fly ash in concrete. [7]
(b) What is M-sand ? Discuss about its advantages and disadvantages. [7]

[3½ x 4=14]

Q.4 Write short notes on:
(a) recycled aggregate
(b) Light weight aggregate
(c) Durability of concrete
(d) Sulphate attack

Q.5 (a) Mention the materials used in Self-compacting concrete. State advantages and disadvantages of self-compacting concrete. [7]
(b) Briefly discuss the advantages and disadvantages of Ready-mix concrete [7]

Q.6 (a) What is durability of concrete? what are the factors affecting durability of concrete [7]
(b) What is maturity of concrete? Explain its significance in gaining the strength of concrete. [7]

Q.7 Explain significance of concrete mix design and write the steps involved in concrete mix design as per IS code and also discuss the variables in proportioning of concrete [14]

Q.8 Design a concrete mix for grade M 25
a. Grade designation: M 25
b. Type of cement: OPC 43 grade
c. Max. nominal size of aggregates 20mm down
d. Min cement content: 300kg/m3
e. Water cement ratio :0.5
f. Workability: 75mm slump
g. No chemical admixture
h. Fine aggregate: zone II
i. Exposure condition: moderate
j. Method of concrete placing: manual
k. Max cement content: 450kg/ m3
l. Specific gravity of cement: 3.15
m. Specific gravity of coarse aggregate :2.80
n. Water absorption of coarse aggregate :1%
o. Free surface moisture: nil
p. Specific gravity of fine aggregate :2.65
q. Water absorption of fine aggregate: 2%
r. Free surface moisture: 2% [14]

Q.9 (a) Define shrinkage and creep of concrete? Discuss about the factors affecting shrinkage of concrete. [7]
(b) Mention Non-destructive tests conducted on hardened concrete. Explain any two of them. [7]

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